

Q1 -- Name:

2pt

Question 1 Are these vectors orthogonal?

$$\vec{u} = \begin{bmatrix} 2 \\ -3 \end{bmatrix}, \quad \vec{v} = \begin{bmatrix} 6 \\ 4 \end{bmatrix}$$

Are these vectors similar? Why/why not? Write a line expression that would separate these points.

2pt

Question 2 A linear model is $y = 2x_0 - x_1 + b$ What is the value of b so that the decision boundary passes through both points (1,4) and (3,2)?

2pt

Question 3 True or False (circle one and briefly explain): T / F: In supervised machine learning, we need training data with known outputs (ground truth) to estimate model parameters. Explanation:

2pt

Question 4 The equation $3x + 4y = 12$ represents a line in 2D.

a) Write this in vector form $\vec{n}^T \vec{x} = d$, identifying the normal vector $\vec{n}$.

b) What is the unit normal vector $\hat{n}$?

1pt

Question 5 Which of the following best describes what a deep neural network does geometrically? (Circle one)

1. Applies a single linear transformation to map inputs to outputs
2. Applies multiple nonlinear transformations to map input points to an output space where classification is easier
3. Simply memorizes the training data
4. Creates a lookup table from inputs to outputs

1pt

Question 6 If $\lambda = 3$ is an eigenvalue of matrix B with eigenvector $\vec{e} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ compute $B\vec{e}$.